Elementary Analysis

Section 30: L'Hospital's Rule

Exercises 30.1–30.7, with every subpart
Detailed worked solutions

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Sections 21–38. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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30 L'Hospital's rule

L'Hospital's rule compares a quotient with a quotient of derivatives; it does not say that every limit may be evaluated by differentiating. In the 0/0 case, both numerator and denominator must tend to zero. Ross also permits the case in which the absolute value of the denominator tends to infinity, without requiring a numerator limit. In either case the functions must be differentiable on the relevant deleted neighborhood, the derivative of the denominator must be nonzero there, and the derivative quotient must have a limit. That limit may be finite or infinite.

For positive quantities with variable exponents, take a logarithm first. If $A(x) > 0$, then $A(x)^{B(x)} = \exp(B(x) \log A(x))$; after finding the exponent's limit, use continuity of the exponential.

30.1 Exercise 30.1: Four introductory limits

What we must show. Evaluate each limit and justify the method, using the actual numerator and denominator in the printed exercise.

(a) The quotient $(e^{2x} - \cos x)/x$ at zero.

Both numerator and denominator tend to zero, they are differentiable, and the denominator derivative is $1 \neq 0$. L'Hospital's rule therefore gives

$$\lim_{x \to 0} \frac{e^{2x} - \cos x}{x} = \lim_{x \to 0} (2e^{2x} + \sin x) = 2.$$

(b) The quotient $(1 - \cos x)/x^2$ at zero.

The original form is 0/0, and the derivative of x^2 is nonzero for $x \neq 0$. The derivative quotient has the known limit

$$\lim_{x \to 0} \frac{\sin x}{2x} = \frac{1}{2}.$$

L'Hospital's rule thus gives 1/2 for the original limit. The last step uses the basic sine limit, not another unjustified cancellation at zero.

(c) The quotient x^3/e^{2x} at positive infinity.

At every differentiation stage the exponential denominator tends to infinity and has nonzero derivative. Successive derivative quotients reduce the polynomial degree:

$$\frac{x^3}{e^{2x}}, \quad \frac{3x^2}{2e^{2x}}, \quad \frac{6x}{4e^{2x}}, \quad \frac{6}{8e^{2x}}.$$

The final quotient tends to zero. Applying the rule backward through these three stages proves that the original limit is 0.

(d) A difference of square roots at zero.

For $0 < |x| < 1$, multiply by the conjugate:

$$\frac{\sqrt{1+x} - \sqrt{1-x}}{x} = \frac{(1+x) - (1-x)}{x(\sqrt{1+x} + \sqrt{1-x})} = \frac{2}{\sqrt{1+x} + \sqrt{1-x}} \longrightarrow 1.$$

This direct algebraic method is shorter than L'Hospital's rule and also keeps the real domain explicit.

Conclusion. The four limits are 2, 1/2, 0, and 1.

30.2 Exercise 30.2: Cancellation, repeated differentiation, and a variable power

What we must show. Evaluate the four limits, checking why each reduction is valid.

(a) The quotient $x^3/(\sin x - x)$.

The numerator and denominator and their first two derivative pairs tend to zero. Differentiating each pair produces

$$\frac{x^3}{\sin x - x}, \quad \frac{3x^2}{\cos x - 1}, \quad \frac{6x}{-\sin x}, \quad \frac{6}{-\cos x}.$$

On a sufficiently small deleted neighborhood of zero, each denominator derivative used in passing to the next pair is nonzero: these are $\cos x - 1$, $-\sin x$, and $-\cos x$. The final quotient tends to -6, so three valid applications of L'Hospital's rule yield -6.

(b) The quotient $(\tan x - x)/x^3$.

This is $0/0$ near zero, where tangent is defined. The denominator derivative $3x^2$ is nonzero for $x \neq 0$. The derivative quotient is

$$\frac{\sec^2 x - 1}{3x^2} = \frac{1}{3}\left(\frac{\tan x}{x}\right)^2 \longrightarrow \frac{1}{3},$$

because $\tan x/x = (\sin x/x)/\cos x \to 1$. Thus the requested limit is $1/3$.

(c) The difference $1/\sin x - 1/x$.

Combine the fractions before applying any limit rule:

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}.$$

The first derivative quotient is $(1 - \cos x)/(\sin x + x\cos x)$, again of form $0/0$. Its derivative quotient is

$$\frac{\sin x}{2\cos x - x \sin x} \longrightarrow 0.$$

The final denominator tends to 2 and hence is nonzero near zero. The intermediate denominator $\sin x + x\cos x$ is nonzero for sufficiently small nonzero x, because its ratio to x tends to 2. Thus both uses of L'Hospital's rule are legitimate and the original limit is 0.

(d) The expression $(\cos x)^{1/x^2}$.

For $|x| < \pi/2$, the base is positive, so its logarithm is $\log(\cos x)/x^2$. This is $0/0$, and its derivative quotient is

$$\frac{-\tan x}{2x} \longrightarrow -\frac{1}{2}.$$

L'Hospital's rule gives the logarithmic limit $-1/2$. Continuity of the exponential now gives the original limit $e^{-1/2}$.

Conclusion. The answers are -6, $1/3$, 0, and $e^{-1/2}$.

30.3 Exercise 30.3: Recognizing when differentiation is unnecessary

What we must show. Evaluate the four limits, including a correction to a printed selected answer.

(a) The quotient $(x - \sin x)/x$ at positive infinity.

For the expression printed in the exercise,

$$\frac{x - \sin x}{x} = 1 - \frac{\sin x}{x}, \qquad \left|\frac{\sin x}{x}\right| \le \frac{1}{x} \longrightarrow 0.$$

The limit is therefore 1. The selected-answer entry 0 in the linked edition is inconsistent with this printed expression. Differentiating would give $1 - \cos x$, which has no limit; this simply means L'Hospital's rule is not useful here. Its failure to supply a derivative-quotient limit says nothing against the elementary estimate above.

(b) The power $x^{\sin(1/x)}$ at positive infinity.

Its logarithm is

$$\sin(1/x) \log x = \frac{\sin(1/x)}{1/x} \frac{\log x}{x}.$$

The first factor tends to 1. The second tends to 0 by L'Hospital's rule, since $(\log x)'/x' = 1/x \to 0$ and $x \to \infty$. Thus the logarithm tends to zero and the power tends to $e^0 = 1$.

(c) The quotient $(1 + \cos x)/(e^x - 1)$ as $x \to 0^+$.

The numerator tends to 2, while the denominator tends to zero through positive values. Consequently the limit is $+\infty$. Explicitly, near zero the numerator is greater than 1. For any prescribed $M > 0$, continuity of e^x makes $0 < e^x - 1 < 1/M$ for sufficiently small $x > 0$, giving the quotient greater than M. This is not a 0/0 situation.

(d) The quotient $(1 - \cos 2x - 2x^2)/x^4$ at zero.

For transparency, list all derivatives through the first nonzero limiting values:

order	numerator derivative	denominator derivative
0	$1 - \cos 2x - 2x^2$	x^4
1	$2 \sin 2x - 4x$	$4x^3$
2	$4 \cos 2x - 4$	$12x^2$
3	$-8 \sin 2x$	$24x$
4	$-16 \cos 2x$	24

At orders 0 through 3 both entries tend to zero. Every denominator derivative required by the rule is nonzero for nonzero x near zero. The last ratio tends to $-16/24 = -2/3$. Four applications, read backward, prove that this is the requested limit.

Conclusion. For the printed questions, the limits are 1, 1, $+\infty$, and $-2/3$.

30.4 Exercise 30.4: Changing a right-hand zero limit into an infinite limit

What we must show. For $g(y) = f(1/y)$, prove equivalence of the two limits, including infinite limits.

Plan. The substitution is reversible: $y \to +\infty$ corresponds exactly to $x = 1/y \to 0^+$.

From a right-hand limit of f to a limit of g.

Suppose first $f(x) \to L \in \mathbb{R}$ as $x \to 0^+$. Given $\varepsilon > 0$, choose $\delta > 0$, with $\delta < a$ if a is finite, such that $0 < x < \delta$ implies $|f(x) - L| < \varepsilon$. If $y > 1/\delta$, then $0 < 1/y < \delta$, so $|g(y) - L| < \varepsilon$. This proves $g(y) \to L$ as $y \to \infty$.

Reverse the argument.

Suppose $g(y) \to L \in \mathbb{R}$. Choose $Y > 0$ inside its domain such that $y > Y$ implies $|g(y) - L| < \varepsilon$. Let $\delta > 0$ be at most $1/Y$ and small enough that $(0, \delta) \subset (0, a)$. Then $0 < x < \delta$ implies $1/x > Y$, and

$$|f(x) - L| = |g(1/x) - L| < \varepsilon.$$

Thus $f(x) \to L$ as $x \to 0^+$.

Use the same substitution for infinite limits.

For a limit $+\infty$, replace $|f(x) - L| < \varepsilon$ and $|g(y) - L| < \varepsilon$ by $f(x) > M$ and $g(y) > M$, respectively, for arbitrary real thresholds M. The same reciprocal choices of Y and δ give both implications. For $-\infty$, use the inequalities $f(x) < M$ and $g(y) < M$. No other part of the proof changes.

Conclusion. Either limit exists exactly when the other does, and their finite or infinite values are equal. The convention $a^{-1} = 0$ when $a = \infty$ gives precisely the correct transformed domain.

30.5 Exercise 30.5: Three exponential limits

What we must show. Evaluate the limits by taking logarithms on domains where the bases are positive.

(a) The expression $(1 + 2x)^{1/x}$ as $x \to 0$.

For $|x| < 1/2$, the base is positive. Its logarithm is $\log(1 + 2x)/x$, of form $0/0$. The denominator derivative is 1, and the numerator derivative is $2/(1 + 2x) \to 2$. Thus the logarithm tends to 2, and the original expression tends to e^2.

(b) The expression $(1 + 2/y)^y$ as $y \to \infty$.

Set $x = 1/y \to 0^+$. The expression becomes $(1 + 2x)^{1/x}$, whose right-hand limit is e^2 by part (a). Exercise 30.4 rigorously justifies the reciprocal change of variable.

(c) The expression $(e^x + x)^{1/x}$ as $x \to \infty$.

The base is positive for $x > 0$. Factor out e^x inside its logarithm:

$$\frac{\log(e^x + x)}{x} = \frac{x + \log(1 + xe^{-x})}{x} = 1 + \frac{\log(1 + xe^{-x})}{x}.$$

L'Hospital's rule gives $x/e^x \to 0$, so $\log(1 + xe^{-x}) \to 0$. Dividing this term by $x \to \infty$ still gives zero; for example its absolute value is eventually at most $1/x$. The logarithmic limit is 1, and the original limit is e.

Conclusion. The three limits are e^2, e^2, and e.

30.6 Exercise 30.6: Recovering limits of a function and its derivative from their sum

What we must show. Given $f(x) + f'(x) \to L \in \mathbb{R}$ at positive infinity, prove $f(x) \to L$ and $f'(x) \to 0$.

Plan. Multiplication by e^x makes the sum appear as a product derivative.

Apply the denominator-infinity version of the rule.

Write

$$f(x) = \frac{f(x)e^x}{e^x}.$$

Both numerator and denominator are differentiable on (c, ∞), the denominator tends to $+\infty$, and its derivative e^x never vanishes. The derivative quotient is

$$\frac{(f(x)e^x)'}{(e^x)'} = \frac{e^x(f'(x) + f(x))}{e^x} = f'(x) + f(x) \longrightarrow L.$$

Theorem 30.2 therefore gives $f(x) \to L$. It is unnecessary to know whether the numerator itself has a limit: Ross's denominator-infinity formulation does not require that.

Subtract two now-known finite limits.

Since $f'(x) = [f(x) + f'(x)] - f(x)$, ordinary limit arithmetic gives $f'(x) \to L - L = 0$.

Conclusion. Both conclusions follow without assuming a limit of f or f' at the outset.

30.7 Exercise 30.7: Why zeros of the denominator derivative cannot be canceled away

What we must show. Analyze $f(x) = x + \cos x \sin x$ and $g(x) = e^{\sin x} f(x)$ and explain the apparent L'Hospital paradox.

(a) Both functions tend to positive infinity.

Since $|\cos x \sin x| \le 1$, we have $f(x) \ge x - 1 \to \infty$. Also $e^{\sin x} \ge e^{-1} > 0$, so for $x > 1$,

$$g(x) \ge e^{-1}(x - 1) \longrightarrow \infty.$$

(b) Differentiate and factor.

Using $1 = \sin^2 x + \cos^2 x$,

$$f'(x) = 1 - \sin^2 x + \cos^2 x = 2\cos^2 x.$$

The product and chain rules give

$$\begin{aligned} g'(x) &= e^{\sin x} \cos x \, f(x) + e^{\sin x} \, 2\cos^2 x \\ &= e^{\sin x} \cos x [2\cos x + f(x)]. \end{aligned}$$

(c) Cancel only at points where cancellation is permitted.

If $x > 3$, then $2\cos x + f(x) \geq -2 + x - 1 = x - 3 > 0$. When also $\cos x \neq 0$, the denominator derivative is nonzero and

$$\frac{f'(x)}{g'(x)} = \frac{2e^{-\sin x}\cos x}{2\cos x + f(x)}.$$

If $\cos x = 0$, however, both derivatives vanish and the left side is undefined. The algebra does not define this original derivative quotient at those points.

(d) Distinguish a continuous extension from the actual quotient.

The expression on the right in part (c) is defined for every $x > 3$, and satisfies

$$\left|\frac{2e^{-\sin x}\cos x}{2\cos x + f(x)}\right| \leq \frac{2e}{x-3} \longrightarrow 0.$$

Nevertheless, $g'(\pi/2 + k\pi) = 0$ for arbitrarily large integers k. Thus there is no tail on which g' is everywhere nonzero, as required by L'Hospital's rule. For the original functions, $f(x) > 0$ on $x > 1$, and cancellation of the nonzero $f(x)$ gives

$$\frac{f(x)}{g(x)} = e^{-\sin x}.$$

Along $u_n = \pi/2 + 2\pi n$ this is e^{-1}; along $v_n = 3\pi/2 + 2\pi n$ it is e. Both sequences tend to infinity, and the values have different limits. Therefore f/g has no limit.

Conclusion. The extended derivative expression tends to zero, but the required nonvanishing hypothesis fails. There is no contradiction to L'Hospital's rule.